

# Computational Thinking



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# Introduction

- **A computer is a programmable electronic device that accepts raw data as input and processes it with a set of instructions (a program) to produce the result as output.**
- **It is believed that the Analytical Engine was the first computer, which was invented by Charles Babbage in 1837.**

# Definition of Computer

- **Computers are becoming a necessary component of contemporary society, permeating almost every area of our daily life.**
- **In order for the computer to perform its functions, a number of hardware and software components must cooperate together.**
- **C – Commonly; O – Operated; M – Machine; P – Particularly; U – Used for; T – Technology; E – Education; R – Research**

# History of Computer

**The first counting device was used by primitive people. They used sticks, stones and bones as counting tools. As the human mind and technology improved with time, more computing devices were developed. Some of the popular computing devices, starting with the first to recent ones, are described below.**

# Abacus



- The history of computers begins with the birth of the abacus, which is believed to be the first computer.
- It is said that the Chinese invented the abacus around 4,000 years ago.

# Napier's Bones



- It was a manually-operated calculating device which was invented by John Napier (1550–1617) of Merchiston. So, the tool became known as “Napier's Bones.”
- It was also the first machine to use the decimal point.

# Pascaline



- Pascaline is also known as the Arithmetic Machine or Adding Machine.
- It was invented between 1642 and 1644 by the French mathematician and philosopher Blaise Pascal.
- It is believed to be the first mechanical and automatic calculator.

# GENERATION

The generation of computers is based on when major technological changes occur within the computer, like the employment of the microprocessor, vacuum tubes, and transistors. The primary generation of this complex system began about 1940, and there are five generations of computers till 2020.

- **FIRST GENERATION (1940–1956)**
- **SECOND GENERATION (1956–1964)**
- **THIRD GENERATION (1964–1971)**
- **FOURTH GENERATION (1972–2010)**
- **FIFTH GENERATION (2010 to Present)**

# Components of Computer System

- **Hardware**
- **Software**
- **Data**
- **Users**

## 1. Hardware

**Hardware consists of the mechanical parts that make up the computer as a machine. The hardware consists of physical devices of the computer. The devices are required for input, output, storage and processing of the data. Keyboard, monitor, hard disk drive, floppy disk drive, printer, processor and motherboard are some of the hardware devices.**

## **2. Software**

**Software is a set of instructions that tells the computer about the tasks to be performed and how these tasks are to be performed. Program is a set of instructions, written in a language understood by the computer, to perform a specific task. A set of programs and documents are collectively called software.**

**The hardware of the computer system cannot perform any task on its own. The hardware needs to be instructed about the task to be performed. Software instructs the computer about the task to be performed. The hardware carries out these tasks. Different software can be loaded on the same hardware to perform different kinds of tasks.**

### **3. Data**

**Data are isolated values or raw facts, which by themselves have no much significance. For example, the data like 20, February, and 1992 just represent values. The data is provided as input to the computer, which is processed to generate some meaningful information. For example, 20, February and 1992 are processed by the computer to give the date of birth of a person.**

### **4. Users**

**Users are people who write computer programs or interact with the computer. They are also known as skinware, liveware, humanware or peopleware. Programmers, data entry operators, system analyst and computer hardware engineers fall into this category.**

# The Input-Process-Output Concept

A computer is an electronic device that: Accepts data; Processes data; Generates output; Stores data. The concept of generating output information from the input data is also referred to as input-process-output concept.

***Input → Process → Output***

The input-process-output concept of the computer is explained as follows:

## ***1. Input***

The computer accepts input data from the user via an input device like keyboard. The input data can be characters, word, text, sound, images, document, etc.

## ***2. Process***

The computer processes the input data. For this, it performs some actions on the data by using the instructions or program given by the user of the data. The action could be an arithmetic or logic calculation, editing, modifying a document, etc. During processing, the data, instructions and output are stored temporarily in the computer's main memory.

## ***3. Output***

The output is the result generated after the processing of data. The output may be in the form of text, sound, image, document, etc. The computer may display the output on a monitor, send output to the printer for printing, play the output, etc.

## ***4. Storage***

The input data, instructions and output are stored permanently in the secondary storage devices like disk or tape. The stored data can be retrieved later, whenever needed.

# Components of Computer Hardware

The computer system hardware comprises of three main components:

- **Input/Output (I/O) Unit**
- **Central Processing Unit (CPU)**
- **Memory Unit**

The I/O unit consists of the input unit and the output unit. CPU performs calculations and processing on the input data, to generate the output. The memory unit is used to store the data, the instructions and the output information.

## **Input/Output (I/O) Unit**

The user interacts with the computer via the I/O unit. The Input unit accepts data from the user and the Output unit provides the processed data, i.e. the information to the user.

# Central Processing Unit

**CPU controls, coordinates and supervises the operations of the computer. It is responsible for processing of the input data. CPU consists of Arithmetic Logic Unit (ALU) and Control Unit (CU).**

- **ALU performs all the arithmetic and logic operations on the input data.**
- **CU controls the overall operations of the computer i.e. it checks the sequence of execution of instructions, and controls and coordinates the overall functioning of the units of computer.**

**Additionally, CPU also has a set of registers for temporary storage of data, instructions, addresses and intermediate results of calculation.**

# Memory Unit

Memory unit stores the data, instructions, intermediate results and output, temporarily, during the processing of data. This memory is also called the main memory or primary memory of the computer. The input data that is to be processed is brought into the main memory before processing. The instructions required for processing of data and any intermediate results are also stored in the main memory. The output is stored in memory before being transferred to the output device. CPU can work with the information stored in the main memory. Another kind of storage unit is also referred to as the secondary memory of the computer. The data, the programs and the output are stored permanently in the storage unit of the computer. Magnetic disks, optical disks and magnetic tapes are examples of secondary memory.

# APPLICATION OF COMPUTERS

Computers have proliferated into various areas of our lives. For a user, computer is a tool that provides the desired information, whenever needed. You may use computer to get information about the reservation of tickets (railways, airplanes and cinema halls), books in a library, medical history of a person, a place in a map, or the dictionary meaning of a word. The information may be presented to you in the form of text, images, video clips, etc.

## *1. Science and Engineering*

Scientists and engineers use computers for performing complex scientific calculations, for designing and making drawings (CAD/CAM applications) and also for simulating and testing the designs. Computers are used for storing the complex data, performing complex calculations and for visualizing 3-dimensional objects. Complex scientific applications like the launch of the rockets, space exploration, etc., are not possible without the computers.

## ***2. Education***

**Computers are extensively used, as a tool and as an aid, for imparting education. Educators use computers to prepare notes and presentations of their lectures. Computers are used to develop computer-based training packages, to provide distance education using the e-learning software, and to conduct online examinations. Researchers use computers to get easy access to conference and journal details and to get global access to the research material.**

### ***3. Medical***

**Medical researchers and practitioners use computers to access information about the advances in medical research or to take opinion of doctors globally. The medical history of patients is stored in the computers. Computers are also an integral part of various kinds of sophisticated medical equipment like ultrasound machine, CAT scan machine, MRI scan machine, etc. Computers also provide assistance to the medical surgeons during critical surgery operations like laparoscopic operations, etc.**

#### ***4. Entertainment***

**Computers have had a major impact on the entertainment industry. The user can download and view movies, play games, chat, book tickets for cinema halls, use multimedia for making movies, incorporate visual and sound effects using computers, etc. The users can also listen to music, download and share music, create music using computers, etc.**

#### ***4. Games***

**A computer can be used to watch a game, view the scores, improve the game, play games (like chess, etc.) and create games. They are also used for the purposes of training players.**

## ***6. Advertising***

**Computers are also used to create an advertisement using the visual and the sound effects. For the advertisers, computer is a medium via which the advertisements can be viewed globally. Web advertising has become a significant factor in the marketing plans of almost all companies. In fact, the business model of Google is mainly dependent on web advertising for generating revenues.**

## ***7. Government***

**The government uses computers to manage its own operations and also for e-governance. The websites of the different government departments provide information to the users. Computers are used for the filing of income tax return, paying taxes, online submission of water and electricity bills, for the access of land record details, etc. The police department uses computers to search for criminals using fingerprint matching, etc.**

## ***8. Home***

**Computers have now become an integral part of home equipment. At home, people use computers to play games, to maintain the home accounts, for communicating with friends and relatives via Internet, for paying bills, for education and learning, etc. Microprocessors are embedded in household utilities like, washing machines, TVs, food processors, home theatres, security devices, etc.**

# Computer Memory

- The computer's memory stores data, instructions required during data processing, and output results.
- Storage may be required for a limited period, instantly, or for an extended period of time.
- Different types of memories are available in a computer, each having its own unique features.
- Cache memory, registers, and RAM are fast memories that temporarily store data and instructions during processing.
- Secondary memory, such as magnetic disks and optical disks, has large storage capacity and stores data and instructions permanently.
- However, secondary memory devices are slower than primary memory.
- Computer memories are organized in a way that provides high performance at minimum cost.

# Memory Unit/Representation

The computer memory stores different kinds of data like input data, output data, intermediate results, etc., and the instructions. Binary digit or bit is the basic unit of memory. A bit is a single binary digit, i.e., 0 or 1. A bit is the smallest unit of representation of data in a computer. However, the data is handled by the computer as a combination of bits. A group of 8 bits form a byte. One byte is the smallest unit of data that is handled by the computer. One byte can store  $2^8$ , i.e., 256 different combinations of bits, and thus can be used to represent 256 different symbols. In a byte, the different combinations of bits fall in the range 00000000 to 11111111.

**A group of bytes can be further combined to form a word. A word can be a group of 2, 4 or 8 bytes.**

***Memory Units:***

**1 bit = 0 or 1**

**1 Byte (B) = 8 bits**

**1 Kilobyte (KB) =  $2^{10}$  = 1024 bytes**

**1 Megabyte (MB) =  $2^{20}$  = 1024 KB**

**1 Gigabyte (GB) =  $2^{30}$  = 1024 MB =  $1024 \times 1024$  KB**

**1 Terabyte (TB) =  $2^{40}$  = 1024 GB =  $1024 \times 1024 \times 1024$  KB**

# Memory Hierarchy

**The memory is characterized on the basis of two key factors—capacity and access time. Capacity is the amount of information (in bits) that a memory can store. Access time is the time interval between the read/write request and the availability of data. The lesser the access time, the faster is the speed of memory. Ideally, we want the memory with fastest speed and largest capacity. However, the cost of fast memory is very high. The computer uses a hierarchy of memory that is organized in a manner to enable the fastest speed and largest capacity of memory.**

# Internal Memory

The key features of internal memory are:

- Limited storage capacity
- Temporary storage
- Fast access
- High cost

Registers, cache memory, and primary memory constitute the internal memory.

The primary memory is further of two kinds—RAM and ROM.

- **Registers are the fastest and the most expensive among all the memory types. The registers are located inside the CPU and are directly accessible by the CPU. The speed of registers is between 1–2 ns (nanosecond). The sum of the size of registers is about 200 B.**
- **Cache memory is next in the hierarchy and is placed between the CPU and the main memory. The speed of cache is between 2–10 ns. The cache size varies between 32 KB to 4 MB.**

- **Any program or data that has to be executed must be brought into RAM from the secondary memory.**
- **Primary memory is relatively slower than the cache memory. The speed of RAM is around 60 ns. The RAM size varies from 512 KB to 3 GB.**

# Secondary Memory

- **The key features of secondary memory storage devices are:**
- **Very high storage capacity**
- **Permanent storage (non-volatile), unless erased by user**
- **Relatively slower access**
- **Stores data and instructions that are not currently being used by CPU but may be required later for processing**
- **Cheapest among all memory**

- To get the fastest speed of memory with largest capacity and least cost, the fast memory is located close to the processor.
- The secondary memory, which is not as fast, is used to store information permanently, and is placed farthest from the processor.

With respect to CPU, the memory is organized as follows:

- *Registers are placed inside the CPU (small capacity, high cost, very high speed)*
- *Cache memory is placed next in the hierarchy (inside and outside the CPU)*
- *Primary memory is placed next in the hierarchy.*
- *Secondary memory is the farthest from CPU (large capacity).*

# CPU REGISTERS

- **Registers are very high-speed storage areas located inside the CPU.**
- **Data and instructions are moved to registers from cache or RAM for processing.**
- **Registers are directly controlled by the Control Unit (CU) during instruction execution.**
- **Therefore, registers are also called the CPU's working memory.**
- **The number and size of registers affect the power and speed of the CPU. A CPU may have 10 to hundreds of registers.**
- **Register size may range from 8 bits to 64 bits.**
- **More registers and larger register size generally improve CPU performance.**

# CACHE MEMORY

- **Cache memory is placed between the CPU and RAM.**
- **It is faster than RAM.**
- **When the CPU needs data or an instruction, it first checks the cache.**
- **If the required information is found in cache, it is called a Cache Hit.**
- **If the information is not found in cache, it is called a Cache Miss.**
- **In case of a cache miss, the required information is retrieved from RAM.**
- **The contents of cache are managed by the Cache Controller, a circuit on the motherboard.**
- **Recently accessed data and instructions help the controller predict what may be needed next.**
- **For good system performance, cache hits should be much more than cache misses.**
- **The two main factors affecting cache performance are: Cache Size, Cache Level (L1, L2 and L3)**

# PRIMARY MEMORY

- Primary memory is the main memory of a computer.
- It is a chip mounted on the motherboard.
- Primary memory is mainly divided into: RAM (Random Access Memory)ROM (Read Only Memory)RAM and ROM – Basic Difference
- *RAM is used for temporary storage of:*
  - Input data
  - Output data
  - Intermediate results

- **Data and instructions required during processing**
- **ROM stores data that generally does not need to be changed.**
- **RAM is volatile, while ROM is non-volatile.**
- **Flash memory is a rewritable form of read-only memory. It is compact, portable, and requires little energy.**

# **RANDOM ACCESS MEMORY (RAM)**

- **RAM stores data and instructions during computer operation.**
- **Data and instructions required by the CPU are first brought into RAM from secondary storage, such as a hard disk.**
- **The CPU interacts with RAM to obtain data and instructions for processing.**
- **RAM is a volatile memory.**
- **It loses information when the computer is switched off.**
- **When the computer is turned on again, required files are loaded from storage into RAM.**

- **Information that needs to be saved for a long time should not be stored only in RAM.**
- **Features of RAM**
- **Provides random access to stored bytes, words, or larger data units.**
- **Any location in RAM can be accessed in approximately the same amount of time.**
- **RAM can be read from and written to.**
- **RAM size is limited because of its high cost.**
- **RAM capacity is generally measured in MB or GB.**

# Read Only Memory

**ROM is a non-volatile primary memory. It keeps its data even when the computer is switched off.**

## ***Main Features of ROM:***

- **Read-only memory:** Data stored in ROM can normally only be read and cannot be easily changed.
- **Permanent storage:** The information stored in ROM remains even when the power is switched off.

- **Programmed by manufacturer:** ROM is generally programmed during manufacturing.
- **Stores startup instructions:** It stores essential programs such as BIOS/firmware needed to start and operate a computer.
- **Device initialization:** ROM stores instructions needed to initialize and check devices connected to the computer.
- **Reliable and Stable :** ROM provides stable storage for instructions that should not be changed frequently

# Secondary Memory

- Secondary memory is used to store data and information permanently.
- It provides large storage capacity.
- It stores data even when the computer is switched off.
- Examples: Hard Disk, SSD, Pen Drive, Memory Card, CD/DVD.

## *Important Features*

**Permanent Storage** – Data remains stored even after power is turned off.

**Large Capacity** – Can store a large amount of data.

**Non-Volatile** – Data is not lost when power is disconnected.

**Low Cost** – Provides more storage at a relatively lower cost.

**Portable** – Devices like pen drives and memory cards can be easily carried.

# Types of Secondary Memory

- **Magnetic Storage:** Hard Disk (HDD)
- **Solid-State Storage:** SSD, Pen Drive, Memory Card
- **Optical Storage:** CD, DVD, Blu-ray Disc